Instrument I1 rerun with a calibrated A/A gate

This report follows up ../weakclass-20261104, where I1 (the character-sum maximum of the factor base's log set) came out **INVALID**. That run's preregistered A/A gate demanded that even and odd null means agree within 5% on every curve, which was too strict for 20 + 20 draws. Protocol protocol.json, WEAKCLASS-I1-20261104, was frozen before data (sha256 dd55966290eb..., commit ddf4a12a).

Changes from the first run

- **Calibrated A/A gate.** Each curve gets 200 null draws in two streams of 100, compared with a two-sample KS test. The gate passes iff the number of curves with p < 0.05 is within the 99th percentile of Binomial(#curves, 0.05). The KS routine was checked on synthetic data before the freeze: 6.0% false rejections in my simulation (5.4% exact; the asymptotic routine is mildly liberal), and 87% power against a 0.5 SD shift.
- **Fresh curves.** Picks are seeded by this protocol's hash. Only 4 of the 54 weak or paired curves coincide with the first run.
- **Out-of-sample overdispersion test.** 30 extra ordinary prime-order curves per size (90 in total) test the first run's descriptive observation that ordinary curves sat in the null's tails too often.
- The curve picking, factor bases, statistic and null law are all unchanged (weakclass.py is a byte copy of the frozen first-run script).

Results (frozen rule)

- **Controls valid.** 7 curves rejected A/A at p < 0.05, against an allowance of 14. The positive control (planted log-AP) has u ≥ 0.95 on 100% of curves.
- **Weak classes: all BLIND.** The mean u is 0.46 for anomalous, 0.44 for small embedding degree and 0.28 for j = 0 with a ζ-stable factor base. No class is consistently beyond the 5–95% band at any size. The j = 0 class sits somewhat low (mean u 0.28, z ≈ −2.3; two curves at u ≤ 0.015 at L = 12, one at 0.03 at L = 14), but its paired ordinary curves are just as low (below), so the shift is not class-specific. The cell rule's "extreme at two sizes" did not replicate.
- **Overdispersion (preregistered, on the 90 independently picked ordinary curves): CALIBRATED.** 11 of 90 fresh ordinary curves are beyond 5%/95% (2 low, 9 high), against 9 expected, with a binomial tail of p = 0.29 (exact expectation 9.85). The first run's excess (10 of 27, P = 0.0035 even after allowing for coarse percentiles) was **not reproduced out of sample**, and it is unexplained. The two runs disagree (Fisher p = 0.008).

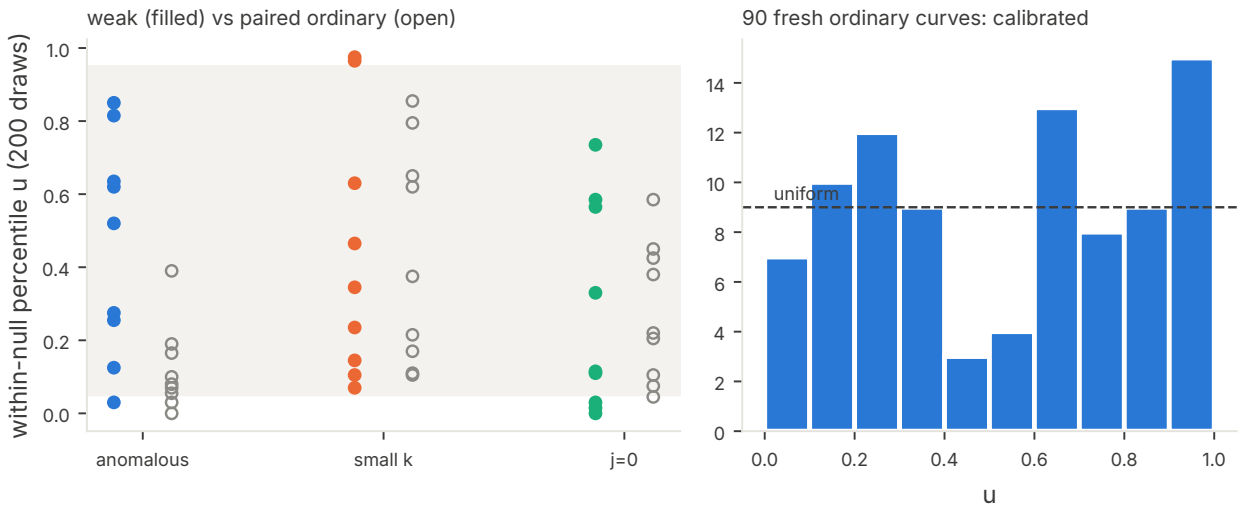

Figure 1: I1 rerun

Unexplained anomaly: the paired ordinary curves sit low (post hoc, found in validation: ../hpseudo-symnull-20261104/VALIDATION-round3.md)

The 27 **paired** ordinary curves in this rerun (class D) have mean u = 0.276 (KS against uniform p = 1.4e-4). The 90 independently picked ordinary curves (class E) have 0.526. Two post-hoc diagnostics follow, labelled as such and not part of the frozen analysis:

- **Fresh null seeds.** Recomputing the D curves' u against 200 freshly seeded null draws gives mean 0.278, with 37% at or below 0.1. The E subset gives 0.541 (7%). The D curves' C values recompute exactly. So it is a property of these curves, not of the null seeds.
- **Same primes, fresh curves.** 23 fresh random ordinary curves on the D curves' primes give mean u 0.484 (4% at or below 0.1). It is not a prime effect.

- The partners of the anomalous curves are lowest, at mean u 0.10 here and 0.29 in the first run. The selection code for D and E is functionally identical (random a, b ≠ 0, prime order, not anomalous, k > 6), and no mechanism has been found. Until it is explained, the paired-difference baseline of the weak-class cells is suspect. The I1 null's calibration is established for independently picked ordinary curves (E) only.

Consequence for the weak-class table

With I1 now valid, **no instrument detects** anomalous, small-embedding-degree or j = 0 curves under the cell rule (3 curves per class per size; all 3 must fall beyond 5–95%). That covers I1 and I2 at p ≤ 2^14 and I3 at p ≈ 2^10 only, where it is rank-saturated. A NULL from any of them excludes only structure of that statistic's own type. It says nothing about whether a curve's DLP is easy. The I1 null of negation-closed random sets is calibrated for independently picked small-x log sets at n ≤ 2^14. The paired-curve anomaly above is open.

Deviations

- No S3 checkpoint (no upload token). With the owner's approval, the pushed branch is the resume record.
- **Storage.** The full rows (585 KB) exceed the 64 KiB in-Git limit even under xz, and this session cannot write to the artifact store. rows-r3.jsonl.xz keeps every field, with the null draws rounded to 3 decimals. analyze_i1.py on it gives a verdict.json identical to the full-precision one. The full file's hash is in data.sha256, and a rerun from the frozen seeds reproduces it exactly.

Reproduce

```
python3 weakclass_i1.py --protocol-sha dd55966290eb11bd7a516640e3438e2f05c7936dd0a07c785f350c720e0b325c --out rows.jsonl
python3 analyze_i1.py rows.jsonl verdict.json      # or: xz -dc rows-r3.jsonl.xz > rows-r3.jsonl; same verdict
python3 figures.py; ./build_pdf.sh
```